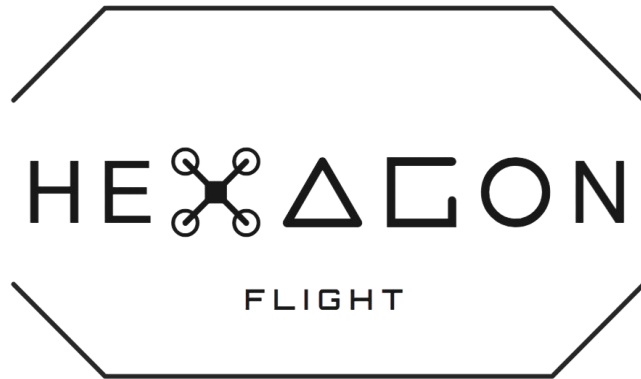




TEAM 03: HEXAGON FLIGHT

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NEED

Project serves as an E-Day presentation for children inspiring them to pursue STEM related degrees.

A reliable method for **autonomous flight** is needed so that drones can safely **navigate indoor obstacle courses** through constrained environments **without** continuous human intervention.



OBJECTIVES

- Comply with UKY and FAA safety requirements.
- Design and develop an interactive drone demonstration for E-Day.
- Showcase principles of robotics and autonomous navigation.
- Inspire student interest in STEM fields or robotic systems.
- Obtain stable flight!



BACKGROUND – EXISTING DEVICE

There is a pre-existing patent device for package delivery:

- City populations are growing exponentially.
- Infrastructure cannot keep up.
- Certain locations are not drivable under certain weather conditions.

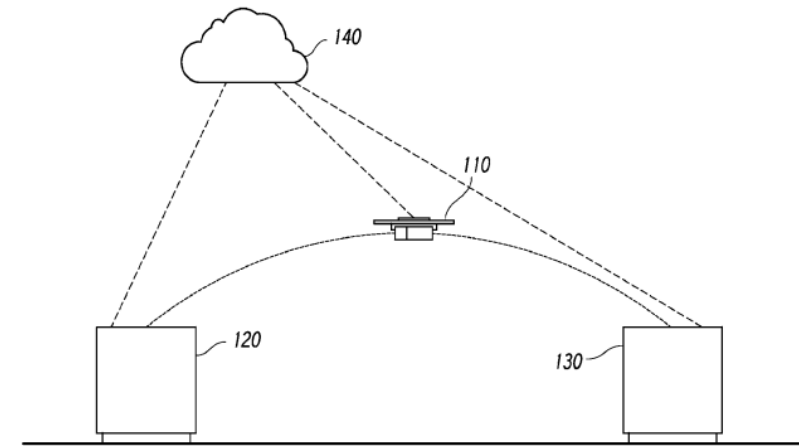


FIG. 1

FIG. 1 shows an embodiment of the disclosed delivery system. The delivery system is comprised of one or more unmanned aerial vehicles (110), ground stations (120 and 130), and a logistics system and network (140). [6]

Our concept is on a low-level scale for recreational use, while current systems are meant for commercial community use.



BACKGROUND – PREVIOUS ATTEMPT

Previous drone is available; various parts were purchased and can be incorporated into design.

After reading through a redacted report following issues were found:

- RAM was not sufficient, leading to reducing various parts for optimization.
- Stabilization was a large issue on why drone was unable to fly.
- As throttle increased, high temperatures and motor loosing were noted.
- Flight was not obtained.

Prospective Plans:

- Search for Flight Controller with more RAM.
- Implement visual markers (like ArUco or AprilTags).
- Propeller guards.

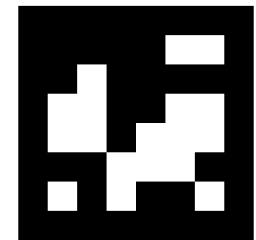
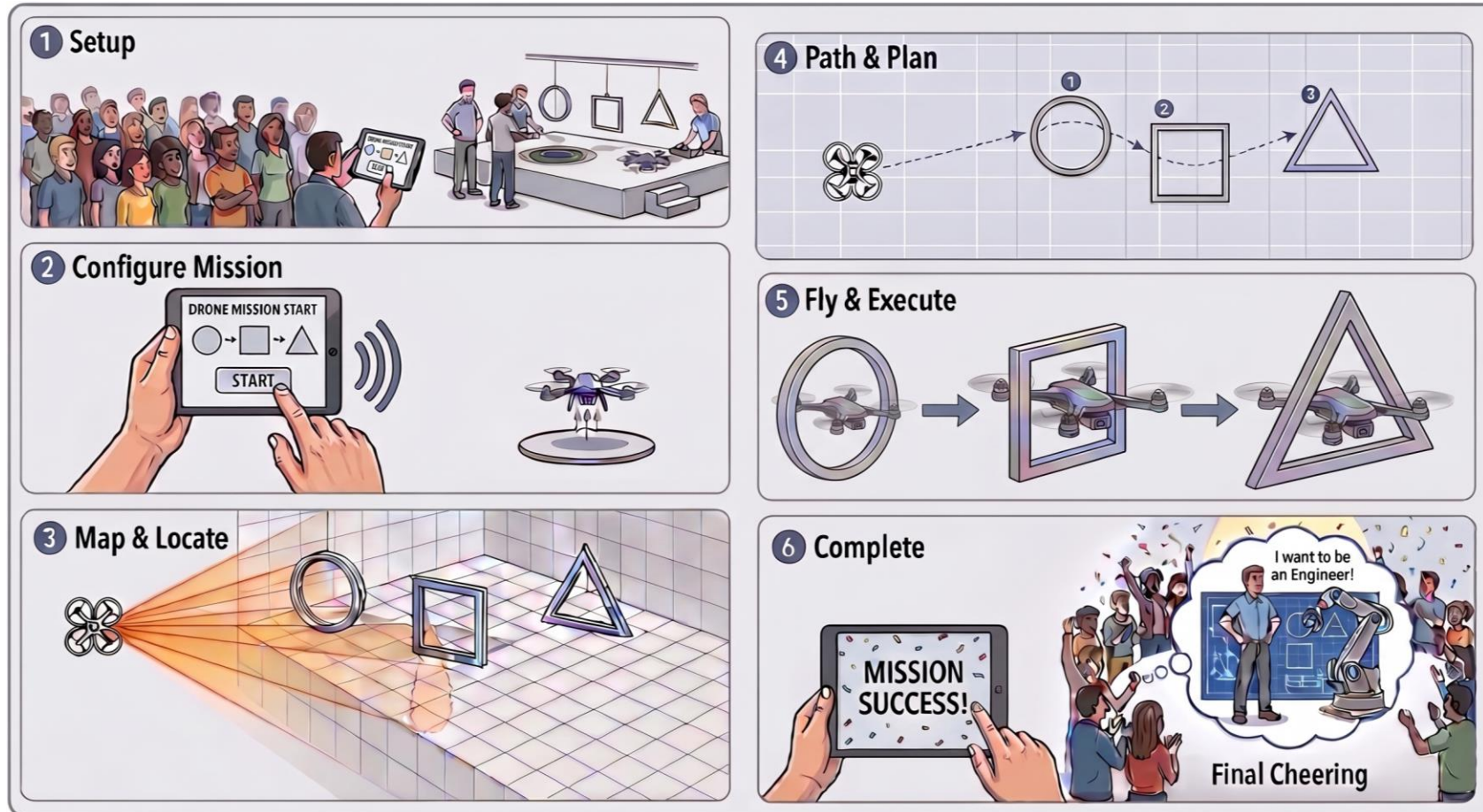


Figure 2: Example of an AprilTag



USER STORY



MARKETING REQUIREMENTS

MR#	Marketing Requirement	Justification
1	The drone shall achieve stable, controlled flight in an enclosed indoor environment	Required for functionality
2	The drone shall execute its entire sequence, from takeoff to landing, autonomously without manual intervention	Fulfills autonomous requirement
3	The drone shall be lightweight (sub-250g)	Project requirement, university and regulatory thresholds
4	The system shall minimize hazard to users or spectators	Essential for indoor flight demonstrations
5	The drone shall detect, identify, and localize the designated geometric goals	Allows for mapping of targets

MARKETING REQUIREMENTS

MR#	Marketing Requirement	Justification
6	The drone shall navigate to and fly through each target goal in the exact sequence designated by the user	Showcases precise path planning and targeted navigation
7	The drone shall have sufficient battery life to complete multiple demonstrations (takeoff, full sequence of goals, and landing) on a single charge	Minimizes battery weight while preserving functionality
8	The system should provide an intuitive user interface that allows the operator to select and submit a sequence of target goals prior to flight	User friendly for everyone
9	The drone should be within budget	Keeps the build financially feasible and accessible
10	The system should be easy and quick to setup and use	Minimizes setup time for faster deployment
11	The system should operate without dependence on external tracking or processing.	Simplifies setup and reduces external dependencies

ENGINEERING REQUIREMENTS 1

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
1	The drone shall be capable of hovering at a specified height without loss of control or drift.	1, 2, 4	Altitude: +/- 10.0 cm; Horizontal drift: <10.0 cm	Stable, reliable flight required to complete task	Test flight, linear distance and height measurement
2	The airframe and mounting fasteners shall maintain structural rigidity and suffer no mechanical deflection or screw loosening under load.	1, 4	0 loose fasteners after 5 minute flight	Repeated demonstrations require reliability and consistency	Test flight and torque/loosening check, marked fasteners
3	The drone shall achieve translational velocity authority in all horizontal directions within the enclosure.	1, 2, 4, 6	Maximum horizontal speed ≥ 0.60 m/s in all cardinal axes	Timely completion of routine	Stopwatch and linear distance, telemetry data

ENGINEERING REQUIREMENTS 2

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
4	The drone's propulsion system shall produce sufficient static thrust to guarantee control authority and rapid disturbance rejection.	1, 2, 4, 6	Thrust-to-Weight Ratio $\geq 2.0:1$ (≥ 500 g total thrust)	Supports stable, safe, consistent flight	Lift force test with mass or force gauge
5	The physical Center of Gravity shall align with the geometric center of the motor thrust plane.	1, 2, 4, 6	CoG offset $\leq \pm 3.0$ mm from thrust center in horizontal axes	Simplifies control parameters, pain point for past groups	CAD analysis, balance test
6	The drone shall execute a controlled vertical descent and upright touchdown upon sequence completion or soft stop.	1, 2, 4, 6, 10	Touchdown vertical speed ≤ 0.25 m/s; Touchdown radius ≤ 0.5 m	Safe and easy-to-use	Test flight landing with vertical speed measurement (telemetry)

ENGINEERING REQUIREMENTS 3

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
7	The drone shall execute the full 3-goal traversal sequence autonomously from takeoff to landing with zero operator inputs.	1, 2, 4, 5, 6, 10	0 user inputs or overrides during sequence, total flight time < 3 minutes	Core project requirement, ease-of-use	Timed autonomous test flight with no human interaction
8	The total mass of the drone, including battery and all payloads, shall not exceed the University and regulatory limit.	3, 10	Target mass <= 245.0g Hard limit at < 250g	Core project requirement, FAA Restriction, University compliance	CAD check, weigh on scale

ENGINEERING REQUIREMENTS 4

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
9	The system shall execute an automatic failsafe landing or motor cut upon loss of companion heartbeat or sensor disconnect.	2, 4	Failsafe trigger response ≤ 200 ms after fault detection	Avoid danger from a rogue drone to spectators and itself	Deliberately disconnect sensor midflight, in a controlled environment
10	Upon operator activation of the UI Emergency Stop button, all motors shall disarm and cease rotation immediately.	4, 8, 10	Button activation to motor disarming response ≤ 150 ms	Need an emergency method to completely stop drone function	Deliberately press UI Emergency Stop button midflight, in a controlled environment
11	Propeller guards shall prevent blade contact with obstacles during low-speed collisions with netting or PVC poles.	1, 4, 10	Blade deflection clearance ≥ 2.0 mm during 0.5 m/s impact	Blade contact with obstacles will destabilize flight	Contact drone with netting midflight at low speed, in a controlled environment

ENGINEERING REQUIREMENTS 5

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
12	All electrical wiring and high-current leads shall be physically contained within the airframe perimeter, away from propeller sweep paths.	1,4	0 mm protrusion into propeller plane; Secured loom	Rotor/wire contact may damage equipment and destabilize flight	Visual inspection of propeller paths before flight
13	Motor surface temperature shall remain within safe thermal margins to prevent magnet degradation.	1,4	Stator/bell surface temperature ≤ 60.0 C	High temperature may damage motors and affect future performance	Measure with infrared temperature gun during and after flight

ENGINEERING REQUIREMENTS 6

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
14	The drone shall identify the 3 target shapes (square, circle, triangle) under standard ambient lighting and accurately gauge their position relative to the drone.	2, 5, 6	$\geq 90\%$ classification accuracy at 0.5 m to 2.0 m distance, position error $\leq \pm 5.0$ cm	Drone must be able to identify targets and its location to them to meet project requirements	Place drone 0.5 – 2 m away from drone and record 10 target identification and position results
15	The drone shall successfully acquire and track target gate contours while in motion.	2, 5, 6	Tracking lock maintained at translational speed ≥ 0.35 m/s	Drone can't reach targets if motion disturbs target tracking information	Confirm tracking lock in hover, provide forward motion, confirm tracking lock maintained

ENGINEERING REQUIREMENTS 7

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
16	The drone shall navigate through the open aperture of each target gate without sustaining structural frame-to-gate collisions.	1, 5, 6	Clearance success rate $\geq$ 80% across consecutive runs	Collision with gates may lead to drone crashing	Repeated test flights to verify 80% success rate over 20 cycles
17	The drone shall support sustained airborne flight on a single fully charged battery pack.	1, 7	Continuous flight time $\geq$ 6.0 minutes with $\geq$ 15% capacity reserve	Inadequate battery life leads to the drone unable to complete its tasks	Power analysis, timed tests
18	The User Interface shall transmit the confirmed goal sequence payload to the SBC and verify receipt acknowledgment.	2, 5, 6, 8	Transmission latency $\leq$ 250 ms; 100% delivery confirmation	Failure to deliver goal sequence means the drone cannot generate or execute a flight path	Network ping

ENGINEERING REQUIREMENTS 8

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
19	The wireless communication link between the User Interface and the drone shall maintain connectivity across the operating radius.	2, 8	Operational range ≥ 10.0 m; Packet loss $< 5\%$	Loss of connectivity can cause drone to lose flight path	Networking test at various ranges
20	The total BoM cost should remain within budget allocations.	2, 9	$\leq \$400.00$ ($\leq 80\%$ of \$500 total budget)	Team budget allowance	Track total costs in BoM
21	The system should support a complete cold-swap battery cycle (including physical pack replacement, FCU calibration, SBC boot, UI reconnection, etc), returning to a ready state within an operational window	4, 10	Ready-to-ready turnaround ≤ 120.0 seconds	Swappable batteries and quick reboot allow for quicker testing and longer flight uptime	Timed swap tests, with experienced and inexperienced users

ENGINEERING REQUIREMENTS 9

ER#	Engineering Requirement	MR#	Metric / Tolerance	Justification	Verification/Testing
22	The drone should maintain body-frame position estimation relative to net origin.	4, 5	Position drift error $\leq \pm 5.0$ cm over 3 minutes	Position tracking of the drone allows for superior 3D mapping	Linear measurement of position drift during test flight
23	Core onboard computing elements shall operate without thermal throttling during active processing.	1, 4	Component temperature ≤ 75.0 deg C under full load	Thermal throttling can lead to drone malfunction	Temperature tests under multiple cycles of vision processing

STANDARDS

- **FAA 14 CFR Part 107: Small UAS Operations:** Operational safety requirements, including visual line of sight, avoiding other aircraft, and preventing careless or reckless operation when applicable.
- **ASTM F3178-24 (UAS Operational Risk Assessment):** Framework for identifying operational hazards and assessing risks before conducting UAS operations and autonomous flight testing.
- **ASTM F2910-22 (Small UAS Design & Construction):** Design, construction, and testing framework for small unmanned aircraft systems.
- **ASTM F3002-22 (sUAS Command & Control):** Framework for designing the command-and-control system, including communications between the aircraft and control system and provisions for loss-of-control/fly-away situations.
- **IEC 62133-2:2017+A1:2021 (Lithium Battery Safety):** Safety requirements and testing for rechargeable lithium cells and batteries under intended use and reasonably foreseeable misuse.



IMPACT STATEMENTS

- **Economic**

- Lightweight design and cost-effective build
- Autonomous operation
- Efficient battery

- **Manufacturability**

- Defined weight and cost constraints
- Modular battery and component design
- Standardized fasteners and frame rigidity
- Simplified setup and intuitive UI

- **Health and Safety**

- Built-in failsafe and emergency stop systems
- Propeller guards and enclosed wiring
- Stable flight and controlled descent
- Thermal limits

- **Social**

- Supports engineering education
- Encourages responsible innovation
- Real-world problem solving



CONCLUSION

- We are developing an autonomous indoor drone capable of identifying and navigating through three designated shapes in a user-selected order. The system will integrate a user interface, computer vision, autonomous navigation, and flight control into one system.
- Our next steps are to continue researching potential hardware and software solutions, finalize the system architecture, and define how each subsystem will communicate.



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QUESTIONS?